Pehchaan

Early diabetes risk screening without a lab test. UnivaBio 2026, theme: AI for Human Health. Muhammad Taqui.

> **Purpose.** Many people with type 2 diabetes do not know they have it, because testing needs a lab visit. Pehchaan answers one question from information a person already knows: should I get a blood test now? It explains every result, works in English and Urdu, and runs fully in the browser with no data sent anywhere.

Key features

- **Six-question screen.** Sex, age, height and weight (live BMI), pregnancies, optional blood pressure, family history.
- **Clear outcome.** Above or below a screening threshold, percentile among study participants, and a specific next step.
- **Exact explanation.** A bar per factor showing its contribution to the score, compared with an average participant.
- **What-if slider.** How the score moves with weight, labelled as a data pattern rather than a promise.
- **Lab result checker.** Fasting, OGTT, HbA1c or random glucose classified by ADA cut-offs; rule-based and separate from the model.
- **Accessible.** Full Urdu interface with right-to-left layout, dark and light themes, mobile layout, printable clinic summary.
- **Private and offline.** Static files only; no server, no account, no storage.
- **Model card in the app.** Data source, metrics with confidence intervals, plots and four stated limitations.

How it works

Logistic regression on the Pima Indians Diabetes Database (768 women), using five features answerable without a blood test. Impossible zeros treated as missing and median-imputed. 25% held out; model choice and threshold (sensitivity target 0.80) fixed by 5-fold cross-validation on the training set; test set used once. The exported model runs in JavaScript, and tests confirm it matches scikit-learn to 1e-9.

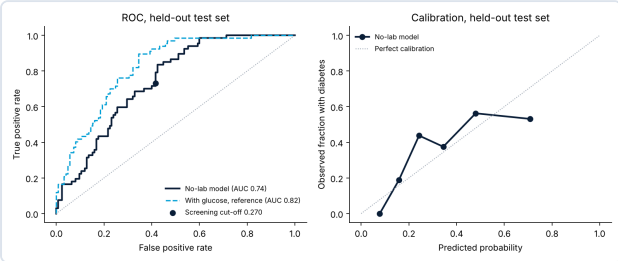
Held-out test results (n = 192, 67 with diabetes)

Metric	Value	95% CI
AUC	0.744	0.67 to 0.81
Sensitivity at 0.27	0.731	0.62 to 0.83
Specificity at 0.27	0.584	0.50 to 0.67

Cross-validated AUC, training set

Logistic regression	0.743
Random forest	0.737
Gradient boosting	0.740
Logistic regression + glucose	0.831

Tree ensembles gave no gain, so the explainable model was kept. Adding glucose lifts test AUC to 0.82, which is why a high score leads to a blood test.



Limitations, stated in the app

Trained only on women of Pima heritage in Arizona; not validated in South Asia, and men receive a warning. Study prevalence was 35%, so raw scores overstate absolute risk. Family history is an approximate mapping. 27% of test-set cases fell below the threshold. Not a medical device.

Finished: everything above. **Planned:** retraining on South Asian data with waist circumference, clinic validation, Android app.

Data: Pima Indians Diabetes Database (NIDDK, via the UCI Machine Learning Repository). Smith et al., Proc. Symp. Computer Applications in Medical Care, 1988, 261 to 265. Code: MIT licence.